
ABSTRACT

The Secure Electronic Health Record (EHR) Management System using Blockchain is designed to address the limitations of traditional healthcare data management systems. Conventional EHR systems face critical challenges such as data fragmentation, lack of interoperability, unauthorized access, and vulnerability to data breaches. These issues reduce trust among stakeholders and hinder efficient sharing of medical information across healthcare providers. The need for a secure, transparent, and tamper-resistant system for managing sensitive patient records has led to the exploration of blockchain technology. This project proposes a blockchain-based EHR system that enhances data integrity and security by leveraging decentralized ledger technology. Blockchain ensures that patient records are stored in an immutable and distributed manner, preventing unauthorized modifications and ensuring data traceability. Each transaction involving patient data is recorded as a block, which is cryptographically secured and linked with previous blocks, thereby improving reliability and trust. The system provides key features including secure patient data storage, role-based access control, data encryption, audit trails, and seamless record sharing among authorized healthcare entities. Patients are given control over their medical data, enabling them to grant or revoke access to healthcare providers. The system architecture ensures scalability, maintainability, and efficient handling of healthcare data transactions. The expected outcomes of this project include improved data security, enhanced privacy, better interoperability between healthcare systems, and increased patient trust. The use of blockchain technology minimizes the risk of data tampering and unauthorized access, leading to a more reliable healthcare information system. Overall, this solution aims to provide a robust and secure platform for modern healthcare data management.

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Chapter-1

INTRODUCTION

Secure Electronic Health Record

Secure Electronic Health Record (EHR) systems play a significant role in modern healthcare by enabling digital storage and management of patient information. However, traditional EHR systems suffer from multiple limitations, including centralized data storage, lack of interoperability, vulnerability to data breaches, and restricted patient control over personal health information. These challenges lead to inefficiencies in data sharing among healthcare providers and increase the risk of unauthorized access or data manipulation. As healthcare data continues to grow in volume and importance, there is a need for a more secure and reliable system that ensures confidentiality, integrity, and availability of sensitive medical records.

To address these challenges, blockchain technology is introduced as a decentralized and tamper-resistant solution for EHR management. Blockchain ensures that data is stored in a distributed ledger, where each transaction is securely recorded and cannot be altered without consensus. This feature enhances transparency, security, and trust among stakeholders. The proposed system combines blockchain with modern web technologies to create a secure platform for managing electronic health records. By integrating features such as role-based access control, encryption, and audit tracking, the system provides improved data security and empowers patients with control over their health data, leading to a more efficient and reliable healthcare information system. The proposed system focuses on improving trust and transparency in healthcare data management through the use of decentralized technologies. It ensures secure and efficient sharing of medical records among healthcare providers while maintaining strict access control. The integration of blockchain with modern web technologies provides a scalable and reliable solution for future healthcare applications.

Chapter-2

GOALS

Objectives (Primary Goals) of the Project

- To design and develop a secure Electronic Health Record (EHR) management system using blockchain technology to ensure data integrity and confidentiality.
- To eliminate the limitations of traditional centralized EHR systems by implementing a decentralized and tamper-proof data storage mechanism.
- To provide role-based access control so that only authorized users such as doctors, patients, and administrators can access specific medical records.
- To enable patients to have full control over their health data, including the ability to grant and revoke access permissions.
- To ensure secure data sharing among healthcare providers while maintaining privacy through encryption techniques.
- To maintain an immutable audit trail of all transactions for transparency and accountability.
- To develop a scalable and efficient system using ASP.NET Core Web API, SQL Server, and IIS for real-time healthcare applications.
- To improve interoperability between different healthcare systems and enhance overall healthcare service quality.

Chapter-3

SOME APPLICATIONS OF SECURE HEALTH RECORD SYSTEM

Healthcare Data Management & Sharing:

1. **Decentralized Data Storage:**
Patient records are stored in a distributed blockchain network instead of a central server, reducing the risk of single-point failure.
2. **Secure Data Sharing:**
Healthcare providers can securely access and share patient records with proper authorization, improving coordination among hospitals.
3. **Patient-Centric Control:**
Patients have full control over their data and can grant or revoke access permissions to doctors and institutions.
4. **Data Integrity Assurance:**
Blockchain ensures that once medical records are added, they cannot be altered, preserving accuracy and reliability.
5. **Transparency and Auditability:**
All record activities are logged, allowing easy auditing and improving trust among stakeholders.

Medical Records Security:

1. **Encryption of Health Data:**
Sensitive patient information is encrypted before being stored, ensuring confidentiality and protection from cyberattacks.
2. **Tamper-Proof Records:**
Blockchain prevents unauthorized modification of records, maintaining data authenticity.
3. **Access Control Mechanism:**
Role-based access ensures only authorized users such as doctors, patients, and admins can

view or update records.

4. Secure Authentication:

Advanced authentication techniques reduce risks of unauthorized login and data breaches.

Hospital & Clinical Management:

1. Doctor / Healthcare Provider:

- Accesses patient records securely
- Updates diagnosis and treatment details
- Shares information with other authorized providers

2. Patient:

- Views personal health records anytime
- Controls access permissions
- Tracks medical history and treatments

3. Administrator:

- Manages system users and roles
- Monitors record access and transactions
- Ensures system compliance and security

4. Blockchain Network:

- Validates and records each transaction
- Maintains a secure and immutable ledger
- Ensures data synchronization across nodes

Chapter-4

SECURE DATA MANAGEMENT AND ACCESS OPTIMIZATION

The concept of secure electronic health record (EHR) management using blockchain ensures safe storage, efficient access, and reliable sharing of medical data among authorized stakeholders. It represents the flow of patient information from patients to healthcare providers, which is then securely recorded in a blockchain ledger and accessed by authorized users. Blockchain technology enhances transparency, security, and traceability by maintaining an immutable record of all transactions. This eliminates issues related to data tampering, unauthorized access, and lack of interoperability in traditional systems. Fig: 4.1 represents the secure EHR management system using blockchain.

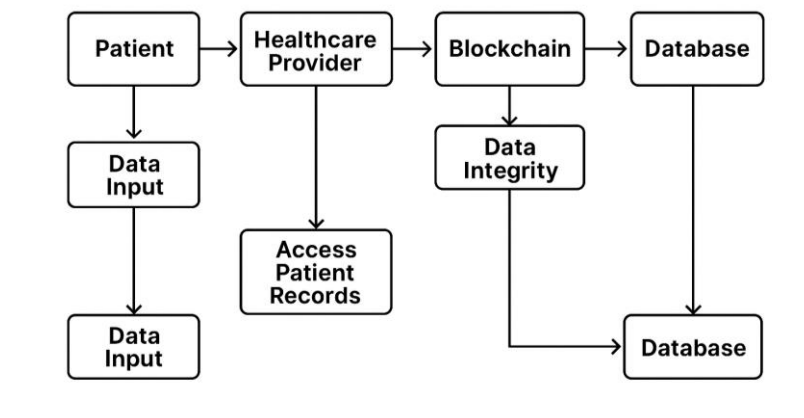


Fig: 4.1 Secure Health Records using Blockchain

1. Secure Data Access and Control

- **How:** Blockchain uses cryptographic keys and role-based access control to manage who can view or update patient records.
- **Benefit:** Ensures only authorized users can access sensitive medical data, improving privacy and security.

2. Real-Time Record Updating and Sharing

- **How:** Healthcare providers update patient records, and transactions are instantly recorded in the blockchain network.
- **Benefit:** Provides up-to-date medical information to authorized entities, improving decision-making and patient care.

3. Immutable Record Management

- **How:** Each medical record is stored as a block linked to previous records, making it tamper-proof.
- **Benefit:** Prevents data alteration and maintains accuracy and trustworthiness of health information.

4. Automated Access using Smart Contracts

- **How:** Smart contracts automatically grant or revoke data access based on predefined conditions set by patients or administrators.
- **Benefit:** Reduces manual intervention, ensures faster data sharing, and minimizes the chances of human error.

5. Interoperability and Data Integration

- **How:** Blockchain enables standardized data exchange between different healthcare systems and hospitals.
- **Benefit:** Improves collaboration among healthcare providers and ensures seamless information flow across platforms.

Chapter-5

SYSTEM DESIGN

System design is a structured approach to organizing system components to ensure scalability, efficiency, security, and reliability. The proposed Secure Electronic Health Record (EHR) Management System is designed by integrating blockchain technology with ASP.NET Core Web API, SQL Server, and IIS. The architecture follows a hybrid model where sensitive healthcare data is securely stored in a centralized database while blockchain is used to maintain integrity and transparency. Fig: 5.1 represents the flow diagram of system design.

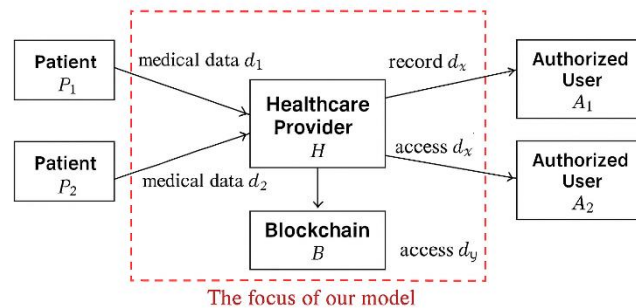


Fig: 5.1 Represent Flow diagram of system design

The system begins with patient registration and data input, where users create accounts and provide their personal and medical information. This data is stored in SQL Server after applying encryption techniques to ensure confidentiality. Each patient is assigned a unique identifier, which acts as a reference for all future transactions. This initial step ensures that every patient record is properly organized and securely maintained within the system. Once registered, healthcare providers interact with the system through a secure login mechanism. Authentication and authorization processes ensure that only permitted users can access sensitive data. Doctors can view patient records after receiving access permission and are allowed to add or update medical details such as diagnosis, prescriptions, and treatment history. Every interaction with patient data is tracked, ensuring accountability and transparency.

Core System Flow

1. Data Creation and Storage

- Patient information is collected and stored in SQL Server.
- Data is encrypted before storage.
- A hash of the record is generated and sent to the blockchain.

2. Access and Update Mechanism

- Healthcare providers request access to patient records.
- Patients grant or deny access based on permissions.
- Approved users can view and update records.

3. Blockchain Recording

- Each transaction is converted into a block.
- Blocks are linked using cryptographic hashing.
- The blockchain ensures immutability and traceability.

The blockchain layer plays a crucial role in maintaining data integrity. Each medical transaction is recorded as a block that is cryptographically secured and linked with previous blocks. This ensures that once data is recorded, it cannot be altered or deleted. A consensus mechanism validates each transaction before adding it to the chain, thereby preventing unauthorized modifications and ensuring data authenticity.

System Components

1. Patient Module:

- Manages patient registration and profile data

- Provides control over data access permissions

2. Healthcare Provider Module:

- Allows doctors to access and update patient records
- Enables secure communication between providers

3. Blockchain Module:

- Stores transaction hashes and logs
- Maintains a decentralized and tamper-proof ledger

4. Database Module (SQL Server):

- Stores actual patient data for fast retrieval
- Works alongside blockchain for hybrid storage

The system follows a hybrid data storage model, which improves both performance and security. While SQL Server handles large volumes of structured data efficiently, blockchain stores only the hash values and transaction logs. This ensures that any attempt to modify data can be easily detected by comparing the stored hash with recalculated values.

Security and Access Optimization

1. Role-Based Access Control

- Different access levels for patients, doctors, and administrators
- Prevents unauthorized data usage

2. Smart Contract Automation

- Automates access permissions and validations
- Reduces manual errors and improves efficiency

3. Audit and Monitoring

- Logs every transaction and access request
- Enables tracking and detection of suspicious activity

The system is deployed using IIS, which ensures reliable hosting and scalability for multiple users. ASP.NET Core Web API enables seamless communication between system components, while the integration of blockchain enhances trust and transparency. Additional auditing features allow administrators to monitor all activities, ensuring compliance with healthcare standards.

The system design provides a secure, scalable, and efficient solution for managing electronic health records. By combining blockchain with modern web technologies, the system ensures improved data protection, controlled access, and reliable healthcare data management.

Chapter-6

DIFFERENT TYPES OF SECURE ELECTRONIC HEALTH RECORD MANAGEMENT USING BLOCKCHAIN

Blockchain Ledger System

A Blockchain Ledger System is a decentralized digital record-keeping mechanism that stores all healthcare transactions across a distributed network. In the context of Electronic Health Records (EHR), it ensures that patient data, medical updates, and access logs are transparent, immutable, and secure. Every transaction related to patient data is recorded in blocks, which are cryptographically linked to form a chain. This prevents unauthorized modification and ensures trust among patients, healthcare providers, and administrators. In healthcare systems, the ledger contains only authorized records such as patient details, diagnosis history, prescriptions, and access logs. Each participant in the network (patients, doctors, and administrators) can view only permitted information based on access rights. Blockchain replaces traditional centralized record storage with a distributed approach, improving data security and availability. Medical records act as digital assets in the ledger, and their lifecycle can be tracked completely. The system ensures that all updates are recorded chronologically, enabling healthcare providers to maintain accurate patient histories and improving decision-making processes.

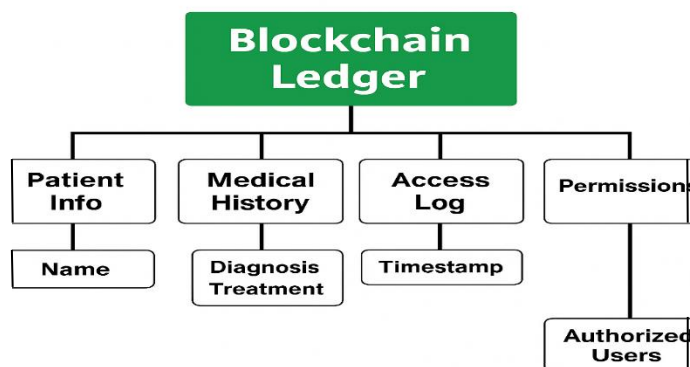


Fig 6.1: Represent the General Ledger structure

Smart Contracts

A Smart Contract is a self-executing program stored on a blockchain that automatically performs actions when predefined conditions are met. In the EHR system, smart contracts are used to manage access permissions, data sharing, and validation of medical records without requiring intermediaries.

Smart contracts function similarly to traditional agreements but are enforced through code. For example, a patient can define conditions under which a doctor can access their records. Once those conditions are satisfied, access is automatically granted without manual approval. This ensures faster, secure, and error-free operations.

Digital Twin Technology (Blockchain-Based Healthcare Model)

A Digital Twin in healthcare is a virtual representation of a patient's medical profile that continuously updates using real-time data. When integrated with blockchain, it ensures that all changes are securely recorded and cannot be altered.

Fig: 6.3 represents the digital twin concept in blockchain-based healthcare systems.

1. Patient Record Creation

A digital profile of the patient is created, including medical history and personal data.

2. Real-Time Data Updates

Medical devices, hospital systems, and healthcare providers continuously update patient data.

3. Blockchain Recording

Each update is recorded on the blockchain as a secure and immutable transaction.

4. Smart Contract Execution

Alerts or actions (e.g., emergency access) are triggered automatically based on predefined conditions.

5. Digital Twin Synchronization

The virtual model reflects the latest patient condition, accessible to authorized users.

Provenance and Traceability in Healthcare Data

Provenance refers to the origin and history of patient data, while traceability is the ability to track all modifications and access events. In blockchain-based EHR systems, provenance ensures that

every medical record can be traced back to its source, such as a specific hospital or doctor.

Blockchain enables:

- Verification of medical record authenticity
- Tracking of all updates and modifications
- Prevention of unauthorized data tampering
- Improved compliance with healthcare regulations

This ensures that healthcare providers rely on accurate and trustworthy patient data.

Blockchain-Based Document Management Systems

A Blockchain-Based Document Management System (DMS) in healthcare securely stores, verifies, and shares patient records such as prescriptions, lab reports, insurance data, and medical certificates. It eliminates issues related to data duplication, loss, and unauthorized modification.

Key processes include:

1. Digital Signature:

Medical records are digitally signed by healthcare providers before being uploaded.

2. Hashing:

A unique hash value (digital fingerprint) of each record is stored on the blockchain, while actual data may be stored in SQL Server or cloud storage.

3. Verification:

Any user can verify the authenticity of records by comparing stored hash values.

4. Smart Contracts:

Automatically manage data access, validation, and sharing processes based on predefined rules.

Chapter-7

Implementation Code For Secure Health Record System

```
import hashlib
import time
import json
import uuid

# Utility Functions

def generate_id(prefix):
    return f'{prefix}_{uuid.uuid4().hex[:8]}'

def current_time():
    return time.ctime()

# User Classes

class User:
    def __init__(self, name, role):
        self.user_id = generate_id(role)
        self.name = name
        self.role = role

class Patient(User):
    def __init__(self, name):
        super().__init__(name, "PATIENT")
        self.records = []
        self.permissions = []
```



```
class Doctor(User):  
    def __init__(self, name):  
        super().__init__(name, "DOCTOR")
```

```
# Smart Contract Simulation
```

```
class AccessContract:  
    def __init__(self, patient_id, doctor_id):  
        self.contract_id = generate_id("CONTRACT")  
        self.patient_id = patient_id  
        self.doctor_id = doctor_id  
        self.active = True  
        self.timestamp = current_time()
```

```
def revoke(self):  
    self.active = False
```

```
# Blockchain Block Structure
```

```
class Block:  
    def __init__(self, index, data, previous_hash):  
        self.index = index  
        self.timestamp = time.time()  
        self.data = data  
        self.previous_hash = previous_hash  
        self.hash = self.compute_hash()
```

```
    def compute_hash(self):  
        block_string = json.dumps({  
"index": self.index,  
        "timestamp": self.timestamp,
```

```
        "data": self.data,
        "prev_hash": self.previous_hash
    }, sort_keys=True)

    return hashlib.sha256(block_string.encode()).hexdigest()
```

Blockchain Ledger

class Blockchain:

```
    def __init__(self):
        self.chain = [self.create_genesis_block()]

    def create_genesis_block(self):
        return Block(0, {"msg": "Genesis Block"}, "0")

    def add_block(self, data):
        prev_block = self.chain[-1]
        new_block = Block(len(self.chain), data, prev_block.hash)
        self.chain.append(new_block)

    def display_chain(self):
        print("\n===== BLOCKCHAIN LEDGER =====")
        for block in self.chain:
            print(f'\nBlock Index: {block.index}')
            print(f'Time: {time.ctime(block.timestamp)}')
            print(f'Data: {block.data}')
            print(f'Hash: {block.hash[:25]}...')
            print(f'Prev Hash: {block.previous_hash[:25]}...')
```

EHR System Core

```
class EHRSystem:
    def __init__(self):
        self.blockchain = Blockchain()
        self.patients = {}
        self.doctors = {}
        self.contracts = []
    # Registration

    def register_patient(self, name):
        patient = Patient(name)
        self.patients[patient.user_id] = patient

        self.blockchain.add_block({
            "action": "REGISTER_PATIENT",
            "patient_id": patient.user_id,
            "name": patient.name
        })

        return patient

    def register_doctor(self, name):
        doctor = Doctor(name)
        self.doctors[doctor.user_id] = doctor

        self.blockchain.add_block({
            "action": "REGISTER_DOCTOR",
            "doctor_id": doctor.user_id,
            "name": doctor.name
        })

        return doctor
    # Permission Control
```

```
def grant_access(self, patient, doctor):
    contract = AccessContract(patient.user_id, doctor.user_id)
    patient.permissions.append(contract)
    self.contracts.append(contract)

    self.blockchain.add_block({
        "action": "GRANT_ACCESS",
        "patient": patient.user_id,
        "doctor": doctor.user_id,
        "contract": contract.contract_id
    })

    print(f'Access granted to {doctor.name}')

def revoke_access(self, patient, doctor):
    for contract in patient.permissions:
        if contract.doctor_id == doctor.user_id:
            contract.revoke()

    self.blockchain.add_block({
        "action": "REVOKE_ACCESS",
        "patient": patient.user_id,
        "doctor": doctor.user_id
    })

    print(f'Access revoked from {doctor.name}')
    return
```

```
# Medical Record Handling
```

```
def add_record(self, doctor, patient, diagnosis, treatment):
    # Check permission
    allowed = False
    for contract in patient.permissions:
        if contract.doctor_id == doctor.user_id and contract.active:
            allowed = True

    if not allowed:
        print("Access Denied!")
        return

    record = {
        "record_id": generate_id("REC"),
        "doctor_id": doctor.user_id,
        "patient_id": patient.user_id,
        "diagnosis": diagnosis,
        "treatment": treatment,
        "timestamp": current_time()
    }

    patient.records.append(record)

    self.blockchain.add_block({
        "action": "ADD_RECORD",
        "record": record
    })

    print("Record added successfully")

def view_records(self, doctor, patient):
    allowed = False
```

```
    for contract in patient.permissions:
        if contract.doctor_id == doctor.user_id and contract.active:
            allowed = True

    if not allowed:
        print("Access Denied!")
        return

    print(f"\nRecords of {patient.name}:")
    for record in patient.records:
        print(record)

# Audit Log Viewer

class Audit:
    def __init__(self, blockchain):
        self.blockchain = blockchain

    def show_audit(self):
        print("\n===== AUDIT LOG =====")
        for block in self.blockchain.chain:
            print(f'Action: {block.data.get('action', 'GENESIS')}')
            print(f'Time: {time.ctime(block.timestamp)}')

# Main Simulation

if __name__ == "__main__":
    system = EHRSystem()

    # Register users
    patient1 = system.register_patient("Nayana")
```

```
doctor1 = system.register_doctor("Dr. Sharma")
```

```
# Grant access
```

```
system.grant_access(patient1, doctor1)
```

```
# Add records
```

```
system.add_record(  
    doctor1, patient1,  
    "Fever",  
    "Paracetamol 500mg"  
)
```

```
system.add_record(  
    doctor1, patient1,  
    "Cold",  
    "Antibiotics"  
)
```

```
# View records
```

```
system.view_records(doctor1, patient1)
```

```
# Revoke access
```

```
system.revoke_access(patient1, doctor1)
```

```
# Try adding record after revoke
```

```
system.add_record(  
    doctor1, patient1,  
    "Follow-up",  
    "Rest"  
)
```

Chapter 8

RESULTS

OUTPUT

B

Block 1 @ Tue Jul 8 15:27:12 2025

Data: {'patient_id': 'PAT_98xy76pq', 'name': 'Nayana', 'action':
'REGISTER_PATIENT'}

Hash: a1f5c3ab879281bb72f0... Prev: 00000000000000000000...

Block 2 @ Tue Jul 8 15:27:13 2025

Data: {'doctor_id': 'DOC_12ab34cd', 'name': 'Dr. Sharma', 'action':
'REGISTER_DOCTOR'}

Hash: b21cbe37bb717aa91f3d... Prev: a1f5c3ab879281bb72f0...

Block 3 @ Tue Jul 8 15:27:15 2025

Data: {'record_id': 'REC_a1b2c3', 'diagnosis': 'Fever', 'treatment': 'Paracetamol',
'status': 'ADDED'} Hash: d41ac78bb91234ef78bd... Prev: c31de98ab8712cc345aa...

Block 4 @ Tue Jul 8 15:27:17 2025 Data: {'action': 'REVOKE_ACCESS', 'patient':
'PAT_98xy76pq', 'doctor': 'DOC_12ab34cd'} Hash: f61ae45cd89012ab56ef... Prev:
e59fd23ab67891bc2345..

Block 5 @ Tue Jul 8 15:27:17 2025 Data: {'action': 'REVOKE_ACCESS', 'patient':
'PAT_98xy76pq', 'doctor': 'DOC_12ab34cd'} Hash: f61ae45cd89012ab56ef... Prev:
e59fd23ab67891bc2345..

OUTPUT EXPLANATION**Block 1**

Component	Description
Patient ID: PAT_98xy76pq	Represents the unique identification number assigned to the patient in the system.
Diagnosis: Fever	Indicates the medical condition identified by the healthcare provider during consultation.
Treatment: Paracetamol 500mg	Specifies the prescribed medication or therapy given to the patient.
Status: “Recorded”	Shows that the medical record has been successfully added to the system.
Timestamp: 1720429632.85	UNIX timestamp representing the exact time of record creation for secure and precise tracking.
Hash: a1f5c3ab87928 1bb72f0...	Unique SHA-256 hash generated for this block to ensure integrity and prevent tampering.
Prev Hash: 00000000000000 000...	Represents the previous block’s hash (Genesis Block), establishing the start of the chain.

Chapter 9

CONCLUSION

The Secure Electronic Health Record (EHR) Management System using Blockchain successfully demonstrates a modern approach to handling sensitive healthcare data with enhanced security, transparency, and reliability. The project addresses the major limitations of traditional EHR systems such as centralized storage, vulnerability to data breaches, lack of interoperability, and limited patient control over personal health information. By integrating blockchain technology, the system ensures that all medical records are stored in a decentralized and tamper-proof manner, thereby improving trust among patients and healthcare providers. The implementation of blockchain provides immutability, meaning that once a record is created, it cannot be altered without proper authorization. This guarantees data integrity and prevents unauthorized modifications. Additionally, the use of smart contract concepts allows automated control over data access, enabling patients to grant or revoke permissions securely. This enhances privacy and ensures that only authorized individuals can access medical records. The use Python and IIS ensures that the system is scalable, efficient, and capable of handling real-world healthcare data requirements. The hybrid storage model, combining blockchain for verification and database systems for performance, provides an optimal balance between security and efficiency. Audit logs further improve accountability by maintaining a complete history of all transactions and actions performed within the system. Overall, the project demonstrates a secure, patient-centric, and future-ready solution for electronic health record management. It highlights the potential of blockchain technology in transforming healthcare systems by ensuring data security, improving interoperability, and enhancing patient trust. The proposed system can be further extended with advanced features such as real-time analytics, integration with IoT health devices, and cloud-based deployment to support large-scale healthcare applications.

Chapter 10

ONLINE CERTIFICATION RELATED TO THE COURSE



||| COURSE COMPLETION CERTIFICATE |||

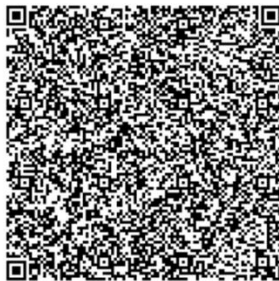
The certificate is awarded to

nayana bashyam

for successfully completing the course

Blockchain Development for Beginners

on June 17, 2026



Issued on: Thursday, June 18, 2026
To verify, scan the QR code at <https://verify.onwingspan.com>



Congratulations! You make us proud!

Satheesha B.N.
Satheesha B. Nanjappa
Senior Vice President and Head
Education, Training and Assessment
Infosys Limited

Chapter 11

PPT PRESENTATION



K. S. INSTITUTE OF TECHNOLOGY
An Autonomous Institution under VTU, Approved by AICTE
Accredited by NBA (CSE & ECE), NAAC with A+ & QS I-GAUGE (GOLD)
No. 14, Raghuvanahalli, Kanakapura Road, Bengaluru – 560109

Department of Computer Science and Engineering
Blockchain Technology(25MCS214A)

Topic: Secure Electronic Health Records

Presented By: B Nayana, 1KS25SCS01 Semester: 2 Mtech, K S Institute of Technology.	Guided By: <u>Dr.Sunita Chalageri</u> , Professor, K S Institute of Technology – Bangalore.
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1



Introduction

- Centralized systems face issues like data breaches, poor interoperability, and limited patient control.
- Growing healthcare data requires secure, reliable, and tamper-proof data management.
- Blockchain provides decentralized, immutable, and transparent data storage for EHR systems.
- Ensures secure access, encryption, audit tracking, and gives patients control over their health data.

3



Objectives

- To design and develop a secure Electronic Health Record (EHR) management system using blockchain technology to ensure data integrity and confidentiality.
- To eliminate the limitations of traditional centralized EHR systems by implementing a decentralized and tamper-proof data storage mechanism.
- To provide role-based access control so that only authorized users such as doctors, patients, and administrators can access specific medical records.
- To enable patients to have full control over their health data, including the ability to grant and revoke access permissions.
- To ensure secure data sharing among healthcare providers while maintaining privacy through encryption techniques.

4



Proposed Methodology

- Collect patient health records from authorized healthcare providers.
- Encrypt all medical data to ensure confidentiality.
- Implement secure user authentication and access control.
- Store records in a secure cloud/database environment.
- Enable authorized sharing of records between healthcare institutions.
- Maintain audit logs to track all access and modifications.
- Provide patients with controlled access to their health records.
- Continuously monitor the system for security threats and data breaches.

Expected Outcome

- Improved data security
- Enhanced patient privacy
- Reduced risk of data breaches
- Faster and secure access to medical records
- Better healthcare data management

Workflow

**Data Collection → Encryption → Secure Storage → Authentication → Authorized Access
→ Audit Monitoring → Secure Sharing**

8



System Architecture

- Patient provides data through a secure data input system
- Healthcare provider receives and accesses patient records
- Data is transmitted to a blockchain network for secure processing
- Blockchain ensures data integrity and tamper-proof storage
- Verified data is stored in a centralized/decentralized database
- Enables secure sharing of medical records
- Improves transparency, trust, and data security
- Reduces risk of data manipulation and unauthorized access

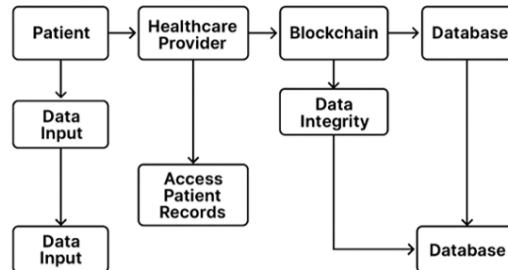


Fig 1: System architecture of Secure health records

13



Implementation

```

class Blockchain:
    def __init__(self):
        self.chain = [self.create_genesis_block()]

    def create_genesis_block(self):
        return Block(0, {"msg": "Genesis Block"}, "0")

    def add_block(self, data):
        prev_block = self.chain[-1]
        new_block = Block(len(self.chain), data,
            prev_block.hash)
        self.chain.append(new_block)

    def display_chain(self):
        print("\n===== BLOCKCHAIN LEDGER =====")
        for block in self.chain:
            print(f"Block Index: {block.index}")
            print(f"Time: {time.ctime(block.timestamp)}")
            print(f"Data: {block.data}")
            print(f"Hash: {block.hash[:25]}...")
            print(f"Prev Hash: {block.previous_hash[:25]}...")

if __name__ == "__main__":
    system = EHRSystem()
    # Register users
    patient1 = system.register_patient("Nayana")
    doctor1 = system.register_doctor("Dr. Sharma")
    # Grant access
    system.grant_access(patient1, doctor1)

    # Add records
    system.add_record(doctor1, patient1, "Fever", "Paracetamol 500mg")
    system.add_record(doctor1, patient1, "Cold", "Antibiotics")

    # View records
    system.view_records(doctor1, patient1)

    # Revoke access
    system.revoke_access(patient1, doctor1)

    # Try adding record after revoke
    system.add_record(doctor1, patient1, "Follow-up", "Rest")
  
```

7



Results and Outputs

Block 1 @ Tue Jul 8 15:27:12 2025

Data: {'patient_id': 'PAT_98xy76pq', 'name': 'Nayana', 'action': 'REGISTER_PATIENT'}

Hash: a1f5c3ab879281bb72f0... Prev: 00000000000000000000...

Block 2 @ Tue Jul 8 15:27:13 2025

Data: {'doctor_id': 'DOC_12ab34cd', 'name': 'Dr. Sharma', 'action': 'REGISTER_DOCTOR'}

Hash: b21cbe37bb717aa91f3d... Prev: a1f5c3ab879281bb72f0...

Block 3 @ Tue Jul 8 15:27:14 2025

Data: {'action': 'GRANT_ACCESS', 'patient': 'PAT_98xy76pq', 'doctor': 'DOC_12ab34cd'}

Hash: c31de98ab8712cc345aa... Prev: b21cbe37bb717aa91f3d...

Block 4 @ Tue Jul 8 15:27:15 2025

Data: {'record_id': 'REC_a1b2c3', 'diagnosis': 'Fever', 'treatment': 'Paracetamol', 'status': 'ADDED'} Hash:

d41ac78bb91234ef78bd... Prev: c31de98ab8712cc345aa...

Block 5 @ Tue Jul 8 15:27:16 2025

Data: {'record_id': 'REC_d4e5f6', 'diagnosis': 'Cold', 'treatment': 'Antibiotics', 'status': 'ADDED'} Hash:

e59fd23ab67891bc2345... Prev: d41ac78bb91234ef78bd...



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